

Solving 15-Puzzles Recursively

The algorithm that solves 15-puzzles is similar to the maze-solving algorithm. Each state of the board (that is, one arrangement of tiles) can be thought of as a maze intersection with four hallways to go down. In the case of 15-puzzles, sliding a tile in one of the four directions is like picking a hallway to follow to the next intersection.

Just as you can turn a maze into a DAG, you can convert a 15-puzzle into a tree graph, as in [Figure 12-2](#). The board states are nodes with up to four edges (representing a direction to slide a tile) to other nodes (representing the resultant state). The root node is the starting state of the 15-puzzle. The solved-state node is the one in which the tiles are ordered correctly. The path from the root node to the solved state details the slides needed to solve the puzzle.